THOTA SAI HARSHITH

Mechanical Design Engineer | ReverseEngineering | ProductDevelopment | SOLIDWORKS | CATIA V5

thotasaiharshith@gmail.com +91 9391030680 Linkedin Portfolio India

PROFILE

Mechanical Design Engineer with 1.5+ years of experience in industrial equipment design, reverse engineering, and engineering support for Tata Steel projects. Proficient in Solidworks, Catia V5, AutoCAD, GD&T, Design for Manufacturability (DFM), and Design for Assembly (DFA). Experienced in developing 3D mechanical components and assembly models, manufacturing drawings, fabrication drawings, Bill of Materials (BOM), and engineering documentation. Strong understanding of machine design fundamentals, manufacturing processes, and design validation through site engineering and client collaboration.

EDUCATION

Bachelor of Technology 2021 – 2025
Vishnu Institute of Technology, bhimavaram(Mechanical Engineering) Andhra Pradesh, India
CGPA : 8.52/10

PROFESSIONAL EXPERIENCE

Praxsol Engineering Private Limited Feb/2025– Present
GET (Design Engineer) Andhra Pradesh, India
Key Clients : TATA Steel, MVS ACMEI

Design Engineering Activities

- Developed 200+ production-ready manufacturing, fabrication, and General Assembly drawings for industrial mechanical components and assemblies, supporting production, installation, and maintenance.
- Developed accurate 3D component and assembly models using Solidworks and CATIA V5. Generated and maintained Bill of Materials (BOM), ensuring proper documentation and traceability.
- Applied GD&T, Design for Manufacturability (DFM), and Design for Assembly (DFA) principles to improve manufacturability and assembly feasibility.
- Performed reverse engineering by transforming field measurements, engineering sketches, and existing components into accurate CAD models and manufacturing drawings.

Site Engineering Activities

- Worked on-site at Tata Steel Kalinganagar, supporting design and engineering activities.
- Executed reverse engineering by capturing design intent and critical dimensional data to develop production-ready CAD models and manufacturing drawings.
- Collaborated with clients and plant engineers in technical discussions to deliver engineering solutions.
- Led a team of Graduate Engineer Trainees, ensuring quality and timely delivery of design outputs.
- Ensured accurate alignment of drawings with actual plant conditions through on-site verification, measurements, and installation feasibility checks.

PROJECTS

Reverse Engineering & Product Development Mar/2025 – Present
Tools : Solidworks, CATIA V5, AutoCAD Mechanical

- Executed end-to-end reverse engineering, transforming field measurements, engineering sketches, and existing components into accurate CAD models and manufacturing drawings.
- Developed 3D CAD models and manufacturing drawings for industrial, automobile, and locomotive components, including shafts, gears, gearboxes, pulleys, sprockets, plummer blocks, base frames, casting, sheet metal, and fabricated components.
- Prepared manufacturing, fabrication, and General Arrangement (GA) drawings, ensuring dimensional accuracy, fitment, manufacturability, and installation feasibility.
- Collaborated with clients, plant engineers, and manufacturing teams during technical discussions and engineering reviews while applying GD&T and Design for Manufacturability (DFM) principles.

INTERNSHIP EXPERIENCE:

Duration : June 2024 to July 2024
Origination : SAP Aerospace (IIT JAMMU, Jammu and kashmir)
Designation : Product Development Intern
Duties and Responsibilities:

- Creating the Piping and Instrumentation diagrams of gas turbine test rig.
- Numerical Investigation of gas turbine combustors.
- Research and Development of Nitrogen, Methane and Hydrogen gases.
- Atomization Testing on Different types of Injectors.

SKILLS

Design Softwares

- Solidworks
- CATIA V5
- Autocad Mechanical
- Autocad Plant 3d
- PTC CREO

Design & Engineering

- Reverse Engineering
- 3D CAD Modeling
- Manufacturing & Fabrica Drawings
- General Assembly (GA) Drawings

Standards & Documentation

- GD&T Fundamentals
- Indian Standards (IS)
- American Society of Mechanical Engineers (ASME)
- Bill of Materials (BOM)
- Creation & Documentation
- Design for Manufacturability (DFM)
- Design for Assembly (DFA)

CERTIFICATIONS

- IntroductiontoASME B31.12 Hydrogen Piping and Pipelines
- IDEATION WORKSHOP - “IDEA TO PROTOTYPE”

LANGUAGES

- English
- Telugu
- Hindi

DECLARATION

- I hereby confirm that the information given is true to my knowledge and belief.

STRENGTHS

- Hard Working
- Team Player
- Self Learner
- Self-motivated with a positive attitude

HOBBIES

- Cricket
- Chess
- Travelling

Place: KAKINADA
Date: (THOTA SAI HARSHITH)